GROUPS INVISIBLE TO FINITE-MATRIX LIMITS

SAUERS

Finite-dimensional unitary matrices do not model every countable group in the operator-norm limit. We study this failure through the MF radical: the elements sent to the identity by every homomorphism into every norm matrix corona. We prove that a one-sided self-compression of a property (T) subgroup forces a canonically defined normal subgroup into the MF radical. Normal generation can then force the MF radical to be the whole group. We realize this situation for an elementary group over the binary Leavitt algebra. The resulting group is simple, has property (T), and every homomorphism from it to an MF group is trivial. In particular, not every countable group is MF. Finally, an amalgamated-product construction realizes any chosen countable group as the quotient detected by all homomorphisms to MF groups.

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1. Introduction

A countable group is *MF* if it embeds in the unitary group of a norm matrix corona

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) \Big/ \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \longrightarrow 0\}$$

is the c_0-direct sum. Equivalently, an MF group admits finite-dimensional unitary models whose multiplicative defects tend to zero in operator norm and which asymptotically separate its nonidentity elements. Throughout, *MF group* means this operator-norm notion of Carrión–Dadarlat–Eckhardt and Korchagin [CDE, Kor].

For a countable group G, define its MF radical by

$$\mathrm{Rad}_{\mathrm{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi : G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

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The group G is MF precisely when $\mathrm{Rad}_{\mathrm{MF}}(G)$ is trivial. If $\mathrm{Rad}_{\mathrm{MF}}(G) = G$, every homomorphism from G to an MF group is trivial.

More generally, for $N \trianglelefteq G$ define its *MF closure* by

$$\mathrm{cl}^G_{\mathrm{MF}}(N) = \bigcap\{\ker f : N \leq \ker f,\ f\colon G \to M,\ M\ \mathrm{MF}\}.$$

The image of a corona homomorphism from G is countable and is itself MF; conversely, every MF target embeds in a norm matrix corona. Thus $\mathrm{Rad}_{\mathrm{MF}}(G) = \mathrm{cl}^G_{\mathrm{MF}}(1)$, and G/N is MF precisely when $\mathrm{cl}^G_{\mathrm{MF}}(N) = N$.

Proposition 1.1 (MF residual calculus)**.** *For every countable group G, the subgroup $\mathrm{Rad}_{\mathrm{MF}}(G)$ is fully invariant, the quotient $G/\mathrm{Rad}_{\mathrm{MF}}(G)$ is MF, and*

$$G/N \text{ is MF} \iff \mathrm{cl}^G_{\mathrm{MF}}(N) = N \qquad (N \trianglelefteq G).$$

In particular, G is MF if and only if its MF radical is trivial.

Proof. An intersection of kernels is normal. If α is an endomorphism of G, $x \in \mathrm{Rad}_{\mathrm{MF}}(G)$, and π is a corona homomorphism of G, then $\pi \circ \alpha$ kills x. Hence $\pi(\alpha(x)) = 1$ for every π, which proves full invariance.

Put $R = \mathrm{Rad}_{\mathrm{MF}}(G)$ and enumerate the nonidentity elements of G/R as $(x_j)_{j\geq 1}$, and enumerate the pairs in $(G/R)^2$. For each x_j, the definition of R supplies a corona homomorphism whose value at x_j has positive distance from the identity. Every such homomorphism kills R and therefore descends to G/R. Choose coordinate unitary lifts, using polar correction as in Lemma 4.1. At stage n, take a direct sum of one coordinate from each of the first n models. Choose each coordinate far enough out that the first n multiplication defects are at most $1/n$ and the designated value at x_j retains at least half of its corona-norm separation. Such coordinates exist arbitrarily far out: the defects converge to zero, while the quotient norm is a limsup. The resulting operator-norm asymptotic representation detects every x_j, so its corona homomorphism is faithful on G/R. Thus G/R is MF.

Applied to G/N, the same argument says that G/N is MF exactly when the intersection of the kernels of all MF-target maps killing N is N, which is the asserted closure criterion. □

The obstruction is captured by the following criterion.

For a subgroup $L \leq G$, put

$$\mathrm{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}$$

and define its *compression–centralizer defect* by

$$\mathfrak{D}_G(L) = \langle\!\langle [ucu^{-1}, \ell] : u \in \mathrm{Comp}_G(L),\ c \in C_G(L),\ \ell \in L \rangle\!\rangle_G. \tag{1}$$

This subgroup depends only on the pair (G, L).

Theorem A (one-sided compression criterion)**.** *Let G be countable and let $L \leq G$ have property* (T). *If $K \trianglelefteq G$ has property* (T) *and $K \leq \mathfrak{D}_G(L)$, then*

$$K \leq \mathrm{Rad}_{\mathrm{MF}}(G).$$

In particular, a nontrivial such K makes G non-MF; if G has property (T) *and $\mathfrak{D}_G(L) = G$, then* $\mathrm{Rad}_{\mathrm{MF}}(G) = G$*. Every finite-dimensional linear representation of G over every field kills $\mathfrak{D}_G(L)$.*

Corollary 3.4 shows that every generator of $\mathfrak{D}_G(L)$ converges to the identity in normalized Hilbert–Schmidt norm in every operator-norm asymptotic representation. If the image of K in a norm matrix corona were nontrivial, normality would make the projection onto the K-fixed vectors commute with the image of G. Compression to its complementary corner would give another operator-norm asymptotic representation, while the Kazhdan inequality would keep some element of K a fixed positive Hilbert–Schmidt distance from the identity. This contradiction proves the criterion. In finite dimension, Theorem 2.1 proves the corresponding statement over every field without property (T).

Whether every countable group is MF was asked in the group-MF literature and remained open [CDE, Kor, Th, Sh]. For the binary Leavitt algebra, Theorem A answers the question.

Let $L_{\mathbb{F}_2}(1,2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$-algebra generated by s_0, s_1, t_0, t_1 subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{2}$$

For a unital ring R, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group)**.** *Put $R = L_{\mathbb{F}_2}(1,2)$ and $H = \mathrm{EL}_{12}(R)$. Then H is nontrivial, simple, has property* (T)*, and*

$$\mathrm{Rad}_{\mathrm{MF}}(H) = H.$$

Equivalently, every homomorphism from H to an MF group is trivial. In particular, H is non-MF.

The analytic input is Corollary 3.4: under the hypotheses of Theorem A, every operator-norm asymptotic representation (V_n) of G satisfies

$$\left\| V_n([ucu^{-1}, \ell]) - 1 \right\|_2 \longrightarrow 0 \qquad (\ell \in L). \tag{3}$$

Inside H, an explicit element τ satisfies $\tau L \tau^{-1} \leq L$ for the upper-left subgroup $L \cong \mathrm{EL}_3(R)$. The element $c = e_{34}(1)$ centralizes L, and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The element d is nontrivial and H is simple, so d normally generates H. Since $d \in \mathfrak{D}_H(L)$, simplicity first gives $H \leq \mathfrak{D}_H(L)$; thus the compression defect is the ambient Kazhdan group, and Theorem A gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$.

If $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism and V is finite-dimensional, conjugation by $\rho(u)$ maps the commutant of $\rho(L)$ injectively into itself, hence bijectively. Thus ρ kills $\mathfrak{D}_G(L)$. This also shows that the configuration cannot occur in several familiar settings: if G has a faithful finite-dimensional linear representation or is residually finite, then $\mathfrak{D}_G(L) = 1$. If G is amenable, its property-(T) subgroup L is finite [BHV]; then $uLu^{-1} = L$, so ucu^{-1} centralizes L for every $u \in \mathrm{Comp}_G(L)$, and again $\mathfrak{D}_G(L) = 1$.

The argument is specific to operator-norm approximation. It does not prove that H is nonsofic or nonhyperlinear: normalized Hilbert–Schmidt approximations do not provide the operator-norm control used to construct the conjugation representation in Theorem 3.3.

Full MF radicals determine prescribed MF quotients as follows.

Theorem C (prescribed MF quotients)**.** *Let B be a countable group with $\mathrm{Rad}_{\mathrm{MF}}(B) = B$, and let $1 \neq d \in B$ normally generate B. Put $A = \langle d \rangle$ and, for a countable group Q, define*

$$W_Q = B *_A (Q \times A).$$

Then the map $\pi_Q \colon W_Q \to Q$ which kills B and projects the second vertex group onto Q is a split epimorphism, and

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

The group W_Q is non-MF. If Q is MF, then

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

Moreover, precomposition with π_Q induces a bijection

$$\mathrm{Hom}(Q, M) \longrightarrow \mathrm{Hom}(W_Q, M)$$

for every MF group M.

Relation to prior work. The group-MF framework used here is developed by Carrión–Dadarlat–Eckhardt and Korchagin [CDE, Kor]; recent permanence results are due to Shulman [Sh]. The operator-norm case remained open after non-approximability had been established for finite Schatten norms [DGLT, LO, Th]. Bachner–Dogon–Lubotzky study an operator–Hilbert–Schmidt stability condition as a possible source of non-MF groups [BDL]. Here the operator norm makes conjugation into a representation in a second norm matrix corona. The Kazhdan projection in that corona then controls the normalized Hilbert–Schmidt commutant, and the normal Kazhdan subgroup argument of Section 4 turns this control into MF-radical membership.

Leavitt introduced the algebras $L_K(1, n)$ as examples of rings with nonunique module rank [Lev]. Their defining relations supply the internal compression used here. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser's normal-subgroup theorem implies simplicity of the elementary group [Pre], while Ershov–Jaikin-Zapirain give property (T) for $\mathrm{EL}_n(R)$ whenever $n \geq 3$ and R is a finitely generated unital associative ring [EJZ, Theorem 1.1]. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

2. One-sided compression in finite dimension

Theorem 2.1 (finite-dimensional commutant rigidity). *Let G be a group, let $L \leq G$, and suppose $u \in G$ satisfies $uLu^{-1} \leq L$. If k is a field, V is a finite-dimensional k-vector space, and $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where $\rho(L)'$ is the commutant of $\rho(L)$ in $\mathrm{End}_k(V)$. Consequently, if $c \in C_G(L)$, then

$$\rho([ucu^{-1}, \ell]) = 1 \qquad (\ell \in L).$$

Proof. Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uhu^{-1} \in L$, then

$$\begin{aligned}\rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u)\\ &= \rho(u)^{-1}x\rho(h')\rho(u)\\ &= \rho(u)^{-1}x\rho(u)\rho(h).\end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and C is finite-dimensional. Its restriction to C is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. It commutes with every $\rho(\ell)$, $\ell \in L$, which proves the last assertion. □

The finite-dimensional hypothesis is essential here. On an infinite-dimensional space an injective endomorphism of the commutant need not be surjective. In Section 3 the Kazhdan projection places the analogous endomorphism inside a norm matrix corona, where stable finiteness supplies the missing surjectivity at the level of fixed-space projections.

3. Kazhdan transport in normalized Hilbert–Schmidt norm

For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \mathrm{tr}_d(a^*a)^{1/2}, \qquad \mathrm{tr}_d = \frac{1}{d}\,\mathrm{Tr}\,.$$

An *operator-norm asymptotic representation* of a countable group G is a sequence of maps $V_n\colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \longrightarrow 0\}.$$

The operator-norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G. Define

$$R_{\infty\to 2}(G) = \bigcap_V K_2(V), \tag{4}$$

where V ranges over all operator-norm asymptotic representations of G. An element is universally Hilbert–Schmidt trivial precisely when it belongs to $R_{\infty\to 2}(G)$.

Lemma 3.1. *For every sequence (m_n), the norm matrix corona*
$$\prod_n M_{m_n}(\mathbb{C})/\bigoplus_n M_{m_n}(\mathbb{C})$$
is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v. Then $x_n^* x_n \to 1$ in norm. For all sufficiently large n, the matrix x_n is invertible, and the unitary $x_n(x_n^* x_n)^{-1/2}$ differs from x_n by $o(1)$. Thus v is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. If A is a stably finite unital C^*-algebra, equivalent projections $p \le q$ satisfy $p = q$: if $w^*w = q$ and $ww^* = p$, then $w \in qAq$ is an isometry in the corner. The corner is finite because $w + (1 - q)$ is an isometry in A. Consequently $ww^* = q$, and hence $p = q$. □

Lemma 3.2 (one-sided order for the Kazhdan projection)**.** *Let L have property* (T)*, let B be a unital C^*-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C^*_{\max}(L) \to B$. If $U \in \mathcal{U}(B)$ satisfies*
$$U\pi(L)U^* \subseteq \pi(L),$$
then
$$U^*PU \le P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\operatorname{Fix}\pi(L)$. If ξ is L-fixed and $\ell \in L$, then
$$\pi(\ell)U^*\xi = U^*\big(U\pi(\ell)U^*\big)\xi = U^*\xi,$$
because $U\pi(\ell)U^*$ belongs to $\pi(L)$. Hence $U^* \operatorname{Fix}\pi(L) \subseteq \operatorname{Fix}\pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B. □

Theorem 3.3 (one-sided Kazhdan transport)**.** *Let $L \le G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \le L$. Let (V_n) be an operator-norm asymptotic representation of G, and let (x_n) be uniformly bounded in operator norm. If*
$$\|V_n(\ell)x_nV_n(\ell)^* - x_n\|_2 \longrightarrow 0 \qquad (\ell \in L),$$
then, for every $\ell \in L$,
$$\|V_n(\ell)V_n(u)x_nV_n(u)^*V_n(\ell)^* - V_n(u)x_nV_n(u)^*\|_2 \longrightarrow 0,$$
$$\|V_n(\ell)V_n(u)^*x_nV_n(u)V_n(\ell)^* - V_n(u)^*x_nV_n(u)\|_2 \longrightarrow 0.$$

Proof. Let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the Hilbert-space ultraproduct $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\mathrm{Ad}(A) - \mathrm{Ad}(B)\| \leq 2\,\|A - B\|\,,$$

so the operator-norm multiplicative defects of (V_n) also make the conjugation actions multiplicative modulo c_0. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\widetilde{\sigma}(g)$ be the class of the coordinate operators $\mathrm{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C}))/\bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each $M_{d_n}(\mathbb{C})$, this is a norm matrix corona with coordinate sizes d_n^2. The preceding estimate shows directly that $g \mapsto \widetilde{\sigma}(g)$ is an exact homomorphism $G \to \mathcal{U}(\mathcal{B})$. Its restriction to L therefore extends to a $*$-homomorphism $C^*_{\max}(L) \to \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection [AW]. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\operatorname{Fix}\sigma(L)$. This representation of $\mathcal{B}$ need not be faithful; it is used only after the following projection identity has been proved inside $\mathcal{B}$. Since $uLu^{-1} \leq L$, the unitary $U = \widetilde{\sigma}(u)$ satisfies the hypothesis of Lemma 3.2, and hence $U^*PU \leq P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 3.1 therefore gives $U^*PU = P$. Hence both U and U^* preserve $\operatorname{Fix}\sigma(L)$, so $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. The two limits in the statement therefore vanish along ω. Since this holds for every free ultrafilter, both sequences converge ordinarily. □

Corollary 3.4. *Let $L \leq G$ have property* (T)*, let $uLu^{-1} \leq L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G) \qquad (\ell \in L).$$

Proof. Let (V_n) be an operator-norm asymptotic representation. If $c \in C_G(L)$, the sequence $(V_n(c))$ asymptotically commutes with $V_n(L)$ in operator norm and hence in normalized Hilbert–Schmidt norm. Theorem 3.3 shows that the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$; explicitly,

$$\left\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\right\| \longrightarrow 0.$$

Therefore

$$\left\|V_n([ucu^{-1}, \ell]) - 1\right\|_2 \longrightarrow 0 \qquad (\ell \in L).$$

Intersecting over (V_n) proves the claim. □

4. The canonical Kazhdan sector and the MF radical

For a normal Kazhdan subgroup $K \trianglelefteq G$, the image of its Kazhdan projection is central relative to every representation of G. Its complement is therefore a canonical invariant sector on which K has no fixed vectors. The finite-dimensional coordinates of this corner carry their own normalized matrix traces. This is the sector on which the Kazhdan inequality detects every nontrivial corona image of K.

Lemma 4.1 (central corona corners)**.** *Let $\rho\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$ and an operator-norm asymptotic representation*

$$W_n\colon G \longrightarrow \mathcal{U}(q_n M_{d_n}(\mathbb{C}) q_n)$$

whose corona class is the corner representation $g \mapsto q\rho(g)$.

Proof. Lift q to projections q_n, and lift each $\rho(g)$ to unitaries $U_n(g)$. Projection lifting follows by spectral rounding. Unitary lifting follows by polar-correcting any lift, since its two unitarity defects converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_n U_n(g) - U_n(g) q_n\| \longrightarrow 0 \qquad (g \in G).$$

Consequently $q_n U_n(g) q_n$, viewed on $q_n \mathbb{C}^{d_n}$, has both unitarity defects converging to zero. Its polar correction $W_n(g)$ is unitary and satisfies

$$\|W_n(g) - q_n U_n(g) q_n\| \longrightarrow 0.$$

The homomorphism relation for ρ now makes (W_n) an operator-norm asymptotic representation. Since $q \neq 0$, infinitely many q_n are nonzero; retain those coordinates. ☐

Theorem 4.2 (normal Kazhdan radical theorem)**.** *Let G be countable, let $D \le R_{\infty\to 2}(G)$, and let $K \trianglelefteq G$ have property* (T) *with $K \le D$. Then*

$$K \le \mathrm{Rad}_{\mathrm{MF}}(G).$$

Proof. Suppose that a homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K. Replace G by $\Theta(G)$, replace K by $\Theta(K)$, and denote the resulting inclusion into the corona again by Θ. The new K is a nontrivial normal property-(T) subgroup. Every one of its elements remains in the universal Hilbert–Schmidt kernel, because any operator-norm asymptotic representation of $\Theta(G)$, precomposed with the quotient map from the original group, is such a representation of the original group.

Let $p_K \in C^*_{\max}(K)$ be the Kazhdan projection and let p be its image in $\mathcal{Q}_{\mathbf{d}}$. Thus p is the projection onto the K-fixed vectors in every faithful representation of the corona. Normality of K gives

$$\Theta(g) p \Theta(g)^* = p \qquad (g \in G).$$

Indeed, in a faithful representation of the corona the two projections have ranges $\Theta(g)\,\mathrm{Fix}\,\Theta(K)$ and $\mathrm{Fix}\,\Theta(K)$; these agree because $gKg^{-1} = K$. Put $q = 1 - p$. The projection q is nonzero, since $p = 1$ would imply that every element of K has trivial image. Lemma 4.1 gives an operator-norm asymptotic representation (W_n) on nonzero corners $q_n M_{d_n}(\mathbb{C}) q_n$.

Choose a finite symmetric Kazhdan set $S \subseteq K$ and a Kazhdan constant $\kappa > 0$. In the corner $q\mathcal{Q}_{\mathbf{d}}q$, the Kazhdan projection is zero. Hence the positive element

$$b = \frac{1}{|S|}\sum_{s\in S}(q\Theta(s)q - q)^*(q\Theta(s)q - q)$$

satisfies

$$b \geq \frac{\kappa^2}{|S|}\, q.$$

Indeed, in every representation of the corner there are no K-fixed vectors, so the defining Kazhdan inequality gives this operator inequality.

The coordinate elements

$$b_n = \frac{1}{|S|}\sum_{s\in S}(W_n(s) - q_n)^*(W_n(s) - q_n)$$

represent b. Taking the normalized trace on $q_n M_{d_n}(\mathbb{C}) q_n$ yields

$$\frac{1}{|S|}\sum_{s\in S}\|W_n(s) - q_n\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

Here the Hilbert–Schmidt norms use the normalized trace of the corner. The order inequality for b says that the negative part of $b_n - (\kappa^2/|S|)q_n$ converges to zero in operator norm, which gives the displayed trace inequality. After passing to a subsequence, one fixed $s_0 \in S$ therefore stays a positive Hilbert–Schmidt distance from the corner identity. This contradicts $s_0 \in K \leq D \leq R_{\infty\to 2}(G)$. Thus every corona homomorphism kills K, as required. □

Proof of Theorem A. For every $u \in \mathrm{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, Corollary 3.4 gives

$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G).$$

Normality of $R_{\infty\to 2}(G)$ therefore gives

$$\mathfrak{D}_G(L) \leq R_{\infty\to 2}(G).$$

Theorem 4.2 now gives $K \leq \mathrm{Rad}_{\mathrm{MF}}(G)$. The two stated consequences follow by taking $K \neq 1$, or $K = G = \mathfrak{D}_G(L)$, respectively.

For a finite-dimensional linear representation over an arbitrary field, Theorem 2.1 kills every generator of $\mathfrak{D}_G(L)$. Its kernel is normal, so it contains $\mathfrak{D}_G(L)$. □

Proposition 4.3 (functoriality and saturation)**.** *Let $L \leq G$ and put $D = \mathfrak{D}_G(L)$.*

If $f\colon G \to Q$ is a homomorphism, then

$$f(D) \le \mathfrak{D}_{f(G)}(f(L)). \tag{5}$$

If $S \le G$ is a nontrivial simple subgroup and $D \cap S \ne 1$, then $S \le D$. If in addition the normal closure of S in G is all of G, then $D = G$.

More generally, suppose f is onto, $S \le D$ is simple, $f(S) \ne 1$, and $f(S)$ normally generates Q. Then

$$\mathfrak{D}_Q(f(L)) = Q.$$

Consequently, if L and G have property (T)*, these hypotheses imply*

$$\mathrm{Rad}_{\mathrm{MF}}(G) = G.$$

Proof. If $u \in \mathrm{Comp}_G(L)$, then $f(u)f(L)f(u)^{-1} \le f(L)$; if $c \in C_G(L)$, then $f(c) \in C_{f(G)}(f(L))$. Therefore f takes every defining generator of D into a defining generator of $\mathfrak{D}_{f(G)}(f(L))$. This proves (5).

The subgroup D is normal in G, so $D \cap S$ is normal in S. Simplicity and $D \cap S \ne 1$ give $D \cap S = S$, hence $S \le D$. Since D is normal in G, it then contains the normal closure of S. Under the second hypothesis this is G.

For the image statement, (5) gives $f(S) \le \mathfrak{D}_Q(f(L))$. The latter subgroup is normal in Q, so it contains the normal closure of $f(S)$ and is therefore all of Q. The last assertion is Theorem A with $K = G$. □

5. The binary Leavitt self-compression

Throughout this section $R = L_{\mathbb{F}_2}(1,2)$ and

$$p = s_0 t_0, \qquad q = s_1 t_1.$$

The relations (2) give

$$p + q = 1, \qquad t_1 q s_1 = 1. \tag{6}$$

In particular $q \ne 0$.

We use indices $0, \dots, 11$ for 12×12 matrices and identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ consisting of the matrices $\mathrm{diag}(A, I_9)$. On $M_3(R)$ consider the unital injective endomorphism

$$\Psi(A) = qI_3 + s_0 A t_0. \tag{7}$$

Here $s_0 A t_0$ denotes the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $q s_0 = t_0 q = 0$ and $t_0 s_0 = 1$ show that Ψ is multiplicative and unital. The identity $t_0 \Psi(A) s_0 = A$ proves injectivity. The term qI_3 is needed for unitality because the map $A \mapsto s_0 A t_0$ sends I_3 to pI_3.

Set

$$X = \begin{pmatrix} s_0 I_3 & s_1 t_0 I_3 \\ 0 & t_1 I_3 \end{pmatrix}, \qquad Y = \begin{pmatrix} t_0 I_3 & 0 \\ s_0 t_1 I_3 & s_1 I_3 \end{pmatrix}.$$

A direct calculation using (2) gives $XY = YX = I_6$. Put

$$\tau = \mathrm{diag}(X, Y) \in \mathrm{GL}_{12}(R). \tag{8}$$

Writing $I = I_6$, the Whitehead factorization is

(9)

$$\operatorname{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix} \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}.$$

Each block-unipotent factor is a product of elementary 12×12 matrices: its off-diagonal 6×6 block may be inserted one entry at a time, and the corresponding matrix units have pairwise zero products. Since $Y = X^{-1}$, (9) proves $\tau \in \mathrm{EL}_{12}(R)$.

Let $H = \mathrm{EL}_{12}(R)$ and let $L = \mathrm{EL}_3(R) \leq H$ be the upper-left corner. Both groups have property (T) by the theorem of Ershov and Jaikin-Zapirain [EJZ, Theorem 1.1]. Block multiplication yields

$$\tau \operatorname{diag}(A, I_9)\, \tau^{-1} = \operatorname{diag}(\Psi(A), I_9). \tag{10}$$

For every $i \neq j$ and $a \in R$, $\Psi(e_{ij}(a)) = e_{ij}(s_0 a t_0)$. Therefore

$$\tau L \tau^{-1} \leq L. \tag{11}$$

The field $\mathbb{F}_2$ is used in the simplicity argument below: its unit group is trivial, so the central subgroup left by the normal-subgroup theorem vanishes. The size 12 records the 3×3 Kazhdan corner and the standard Whitehead doubling above; no minimality assertion is intended.

6. The full MF radical

Proposition 6.1. *The group $H = \mathrm{EL}_{12}(R)$ is simple.*

Proof. The ring $R = L_{\mathbb{F}_2}(1, 2)$ is purely infinite simple [AA] and therefore an exchange ring [Ara]. Let $N \trianglelefteq H$. Preusser's normal-subgroup theorem [Pre, Theorem 3] gives an ideal $I \trianglelefteq R$ such that

$$\mathrm{EL}_{12}(R, I) \leq N \leq C_{12}(R, I),$$

where $C_{12}(R, I)$ is the full congruence subgroup of level I. Since R is simple, either $I = R$ or $I = 0$. In the first case $\mathrm{EL}_{12}(R, I) = H$, so $N = H$. In the second case

$$N \leq C_{12}(R, 0) = Z(\mathrm{GL}_{12}(R)).$$

The center of R is $\mathbb{F}_2$ by [AC, Corollary 4.3]. For completeness, suppose that $z \in Z(\mathrm{GL}_{12}(R))$. Commuting z with the elementary matrices $e_{ij}(1)$ shows that $z = \lambda I_{12}$ for some $\lambda \in R$. Commuting with $e_{ij}(a)$ for arbitrary $a \in R$ then gives $\lambda a = a \lambda$, so $\lambda \in Z(R)^\times = \mathbb{F}_2^\times = \{1\}$. Thus $Z(\mathrm{GL}_{12}(R))$ is trivial, and $N = 1$ when $I = 0$. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. $\square$

Proposition 6.2. *Let*

$$c = e_{34}(1), \qquad \ell = e_{12}(1), \qquad d = [\tau c \tau^{-1}, \ell]$$

in H. Then

$$c \in C_H(L), \qquad d = e_{02}(q) \neq 1, \qquad \langle\!\langle d \rangle\!\rangle_H = H.$$

Proof. Every elementary generator of the upper-left 3×3 corner has both indices in $\{0, 1, 2\}$. The elementary commutator relations show that it commutes with $c = e_{34}(1)$, so $c \in C_H(L)$.

Direct multiplication of the matrices in (8) gives

$$\tau c \tau^{-1} = e_{01}(q) e_{34}(1). \tag{12}$$

The second factor commutes with $\ell = e_{12}(1)$, and $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ gives $d = e_{02}(q)$. This is nontrivial because $q \neq 0$. Finally, d is nontrivial and H is simple by Proposition 6.1, so its normal closure is all of H. □

Proof of Theorem B. The group H is nontrivial and simple by Proposition 6.1. The ring R is finitely generated, and H has property (T) by the Ershov–Jaikin-Zapirain theorem for elementary groups over finitely generated rings [EJZ, Theorem 1.1].

The containment (11), the centrality of c relative to L, and Proposition 6.2 show that $d \in \mathfrak{D}_H(L)$. Since H is simple and $d \neq 1$, Proposition 4.3 gives $\mathfrak{D}_H(L) = H$. The group H has property (T), so Theorem A, applied with $K = H$, gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$.

If M is MF and $f\colon H \to M$ is a homomorphism, compose f with a faithful homomorphism from M to a norm matrix corona. Since $\mathrm{Rad}_{\mathrm{MF}}(H) = H$, this composition and hence f are trivial. If H were MF, its identity homomorphism would contradict this conclusion. □

7. Prescribed MF quotients

Let B be a countable group with $\mathrm{Rad}_{\mathrm{MF}}(B) = B$, and let $1 \neq d \in B$ normally generate B. Put $A = \langle d \rangle$. For a countable group Q, define

$$W_Q = B *_A (Q \times A), \tag{13}$$

where $A \to Q \times A$ is the inclusion into the second factor. The maps from B and $Q \times A$ to Q that are respectively trivial and the projection onto Q agree on A. They therefore define an epimorphism

$$\pi_Q \colon W_Q \longrightarrow Q.$$

It is split by the inclusion $Q \to Q \times A \to W_Q$. The normal-form theorem for amalgamated free products shows that both vertex maps in (13) are injective; in particular, $d \neq 1$ in W_Q.

Proposition 7.1 (universal factorization)**.** *Let T be a group such that every homomorphism $B \to T$ is trivial. Then precomposition with π_Q is a bijection*

$$\mathrm{Hom}(Q, T) \longrightarrow \mathrm{Hom}(W_Q, T).$$

Furthermore,

$$\ker \pi_Q = \langle\!\langle d \rangle\!\rangle_{W_Q}.$$

Proof. Let $f\colon W_Q \to T$ be a homomorphism. Its restriction to B is trivial, so its restriction to the amalgamated subgroup A is trivial. For $(q,a) \in Q \times A$, we then have

$$f(q,a) = f(q,1)f(1,a) = f(q,1).$$

Thus the restriction of f to $Q \times A$ factors uniquely through the projection onto Q. The universal property of the amalgam shows that f factors through π_Q. The factorization is unique because π_Q is surjective.

The element d belongs to $\ker \pi_Q$. Conversely, quotienting W_Q by $\langle\langle d \rangle\rangle_{W_Q}$ kills B, since $\langle\langle d \rangle\rangle_B = B$, and kills the second factor A in $Q \times A$. The resulting quotient is Q, with quotient map induced by π_Q. Hence $\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}$. □

Proof of Theorem C. Every homomorphism from B to an MF group is trivial: compose it with a faithful corona embedding of the target and use $\mathrm{Rad}_{\mathrm{MF}}(B) = B$. Proposition 7.1 therefore gives the asserted bijection for every MF group M. In particular, every homomorphism from W_Q to a norm matrix corona factors uniquely through π_Q. Intersecting their kernels gives

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

The kernel of π_Q belongs to this radical, and it contains the nonidentity element d. Thus W_Q is non-MF. If Q is MF, its MF radical is trivial, and the displayed equality becomes

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

□

The factorization also determines all MF consequences of additional relations. If $N \trianglelefteq W_Q$, then

$$\mathrm{cl}_{\mathrm{MF}}^{W_Q}(N) = \pi_Q^{-1}\left(\mathrm{cl}_{\mathrm{MF}}^{Q}(\pi_Q(N))\right). \tag{14}$$

Indeed, an MF-target homomorphism from W_Q kills N exactly when its unique factor through Q kills $\pi_Q(N)$. Proposition 1.1 and (14) give

$$W_Q/N \text{ is MF} \iff \ker \pi_Q \le N \text{ and } Q/\pi_Q(N) \text{ is MF}.$$

Here, if $\ker \pi_Q \le N$, then $N = \pi_Q^{-1}(\pi_Q(N))$; this is the only additional observation needed for the displayed equivalence.

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